

PSP Oracle Range Partitioning to Hash Partitioning GG Replication POC

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Range to Hash Partitioning

Table DDL

```
CREATE TABLE "PSPADM"."PSP_FINANCIAL_TRANSACTION"
(
  "FINANCIAL_TRANSACTION_SEQ" VARCHAR2(255) CONSTRAINT "SYS_C0078691" NOT NULL ENABLE,
  "VERSION" NUMBER(19,0) CONSTRAINT "SYS_C0078692" NOT NULL ENABLE,
  "CREATOR_ID" VARCHAR2(30),
  "CREATED_DATE" TIMESTAMP (6) CONSTRAINT "SYS_C0078693" NOT NULL ENABLE,
  "MODIFIER_ID" VARCHAR2(30),
  "MODIFIED_DATE" TIMESTAMP (6) CONSTRAINT "SYS_C0078694" NOT NULL ENABLE,
  "REALM_ID" NUMBER(19,0) DEFAULT -1 CONSTRAINT "SYS_C0078695" NOT NULL ENABLE,
  "FINANCIAL_TRANSACTION_AMOUNT" NUMBER(19,4),
  "SETTLEMENT_DATE" TIMESTAMP (6),
  "SETTLEMENT_TYPE_CD" VARCHAR2(255),
  "CREDIT_BANK_ACCOUNT_TYPE" VARCHAR2(255),
  "DEBIT_BANK_ACCOUNT_TYPE" VARCHAR2(255),
  "ON_HOLD" NUMBER(1,0),
  "SKU" VARCHAR2(40),
  "SKU_QUANTITY" NUMBER(10,0),
  "BILLING_DETAIL_FK" VARCHAR2(255),
  "CREDIT_BANK_ACCOUNT_FK" VARCHAR2(255),
  "DEBIT_BANK_ACCOUNT_FK" VARCHAR2(255),
  "COMPANY_FK" VARCHAR2(255),
  "PAYROLL_RUN_FK" VARCHAR2(255),
  "PAYCHECK_SPLIT_FK" VARCHAR2(255),
  "TRANSACTION_TYPE_FK" VARCHAR2(255) CONSTRAINT "SYS_C0078696" NOT NULL ENABLE,
  "LAW_FK" VARCHAR2(255),
  "CURRENT_TRANSACTION_STATE_FK" VARCHAR2(255),
  "ORIGINAL_TRANSACTION_FK" VARCHAR2(255),
  "MONEY_MOVEMENT_TRANSACTION_FK" VARCHAR2(255),
  "ORIGINAL_SETTLEMENT_DATE" TIMESTAMP (6),
  "REFUND_TYPE" VARCHAR2(255 CHAR),
  "COMP_ADJUST_SUBMISSION_FK" VARCHAR2(255 CHAR),
  "TAX_PENALTY_INTEREST_FK" VARCHAR2(255 CHAR),
  "RELATABLE_TRANSACTION_FK" VARCHAR2(255 CHAR),
  "BILL_PAYMENT_SPLIT_FK" VARCHAR2(255 CHAR),
  "COMPANY_LAW_FK" VARCHAR2(255 CHAR),
  "STATUS" VARCHAR2(255 CHAR),
  CONSTRAINT "C_PSP_FINANCIAL_TRANSACTION2" CHECK (DEBIT_BANK_ACCOUNT_TYPE IN('Company', 'Employee',
'Intuit', 'TaxAgency', 'Payee')) ENABLE,
  CONSTRAINT "C_PSP_FINANCIAL_TRANSACTION1" CHECK (CREDIT_BANK_ACCOUNT_TYPE IN('Company', 'Employee',
'Intuit', 'TaxAgency', 'Payee')) ENABLE,
  CONSTRAINT "C_PSP_FINANCIAL_TRANSACTION3" CHECK (REFUND_TYPE IN('ApplyForward', 'Immediate', 'TOR'))
ENABLE,
  CONSTRAINT "C_PSP_FINANCIAL_TRANSACTION0" CHECK (SETTLEMENT_TYPE_CD IN('EFE', 'ACH', 'Wire', 'Cash',
'CheckType', 'Other', 'ApplyForward', 'HPDE', 'TakenOnReturn', 'EFTPS', 'EFTPSDirectDebit', 'EDI')) ENABLE
NOVALIDATE,
  SUPPLEMENTAL LOG GROUP "GGS_90017" ("FINANCIAL_TRANSACTION_SEQ", "REALM_ID") ALWAYS,
  SUPPLEMENTAL LOG DATA (PRIMARY KEY) COLUMNS,
```

```

SUPPLEMENTAL LOG DATA (UNIQUE INDEX) COLUMNS,
SUPPLEMENTAL LOG DATA (FOREIGN KEY) COLUMNS,
SUPPLEMENTAL LOG DATA (ALL) COLUMNS,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK3" FOREIGN KEY ("DEBIT_BANK_ACCOUNT_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_BANK_ACCOUNT" ("BANK_ACCOUNT_SEQ", "REALM_ID") ENABLE NOVALIDATE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK3" FOREIGN KEY ("ORIGINAL_TRANSACTION_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_FINANCIAL_TRANSACTION" ("FINANCIAL_TRANSACTION_SEQ", "REALM_ID") ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK2" FOREIGN KEY ("RELATABLE_TRANSACTION_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_FINANCIAL_TRANSACTION" ("FINANCIAL_TRANSACTION_SEQ", "REALM_ID") ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK11" FOREIGN KEY ("COMP_ADJUST_SUBMISSION_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_COMP_ADJUST_SUBMISSION" ("COMP_ADJUST_SUBMISSION_SEQ", "REALM_ID") ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK8" FOREIGN KEY ("LAW_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_LAW" ("LAW_ID", "REALM_ID") ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK9" FOREIGN KEY ("CURRENT_TRANSACTION_STATE_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_TRANSACTION_STATE" ("TRANSACTION_STATE_CD", "REALM_ID") ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK7" FOREIGN KEY ("TRANSACTION_TYPE_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_TRANSACTION_TYPE" ("TRANSACTION_TYPE_CD", "REALM_ID") ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK14" FOREIGN KEY ("TAX_PENALTY_INTEREST_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_TAX_PENALTY_INTEREST" ("TAX_PENALTY_INTEREST_SEQ", "REALM_ID") ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK5" FOREIGN KEY ("PAYROLL_RUN_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_PAYROLL_RUN" ("PAYROLL_RUN_SEQ", "REALM_ID") ENABLE NOVALIDATE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK4" FOREIGN KEY ("COMPANY_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_COMPANY" ("COMPANY_SEQ", "REALM_ID") ENABLE NOVALIDATE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK2" FOREIGN KEY ("CREDIT_BANK_ACCOUNT_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_BANK_ACCOUNT" ("BANK_ACCOUNT_SEQ", "REALM_ID") ENABLE NOVALIDATE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK10" FOREIGN KEY ("MONEY_MOVEMENT_TRANSACTION_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_MONEY_MOVEMENT_TRANSACTION" ("MONEY_MOVEMENT_TRANSACTION_SEQ", "REALM_ID")
ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK1" FOREIGN KEY ("BILLING_DETAIL_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_BILLING_DETAIL" ("BILLING_DETAIL_SEQ", "REALM_ID") ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK15" FOREIGN KEY ("BILL_PAYMENT_SPLIT_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_BILL_PAYMENT_SPLIT" ("BILL_PAYMENT_SPLIT_SEQ", "REALM_ID") ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK16" FOREIGN KEY ("COMPANY_LAW_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_COMPANY_LAW" ("COMPANY_LAW_SEQ", "REALM_ID") ENABLE,
CONSTRAINT "PSP_FINANCIAL_TRANSACTION_FK6" FOREIGN KEY ("PAYCHECK_SPLIT_FK", "REALM_ID")
REFERENCES "PSPADM"."PSP_PAYCHECK_SPLIT" ("PAYCHECK_SPLIT_SEQ", "REALM_ID") ENABLE
) LOGGING
TABLESPACE "PSP_DATA01"
PARTITION BY RANGE ("SETTLEMENT_DATE")
(PARTITION "FINANCIAL_TXN_MG62008" VALUES LESS THAN (TIMESTAMP' 2009-01-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG12009" VALUES LESS THAN (TIMESTAMP' 2009-03-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG22009" VALUES LESS THAN (TIMESTAMP' 2009-05-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG32009" VALUES LESS THAN (TIMESTAMP' 2009-07-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG42009" VALUES LESS THAN (TIMESTAMP' 2009-09-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG52009" VALUES LESS THAN (TIMESTAMP' 2009-11-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG62009" VALUES LESS THAN (TIMESTAMP' 2010-01-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG12010" VALUES LESS THAN (TIMESTAMP' 2010-03-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG22010" VALUES LESS THAN (TIMESTAMP' 2010-05-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG32010" VALUES LESS THAN (TIMESTAMP' 2010-07-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG42010" VALUES LESS THAN (TIMESTAMP' 2010-09-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG52010" VALUES LESS THAN (TIMESTAMP' 2010-11-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG62010" VALUES LESS THAN (TIMESTAMP' 2011-01-01 00:00:00')
TABLESPACE "PSP_DATA01" ,
PARTITION "FINANCIAL_TXN_MG22021" VALUES LESS THAN (TIMESTAMP' 2021-05-01 00:00:00')
TABLESPACE "PSP_LOB01" ,
PARTITION "FINANCIAL_TXN_MG32021" VALUES LESS THAN (TIMESTAMP' 2021-07-01 00:00:00')
TABLESPACE "PSP_LOB01" ,
PARTITION "FINANCIAL_TXN_MG42021" VALUES LESS THAN (TIMESTAMP' 2021-09-01 00:00:00')
TABLESPACE "PSP_LOB01" ,

```

```

PARTITION "FINANCIAL_TXN_MG52021" VALUES LESS THAN (TIMESTAMP' 2021-11-01 00:00:00')
TABLESPACE "PSP_LOB01" ,
PARTITION "FINANCIAL_TXN_MG62021" VALUES LESS THAN (TIMESTAMP' 2022-01-01 00:00:00')
TABLESPACE "PSP_LOB01" ,
PARTITION "FINANCIAL_TXN_9999" VALUES LESS THAN (MAXVALUE)
TABLESPACE "PSP_LOB01" ) ENABLE ROW MOVEMENT
ALTER TABLE "PSPADM"."PSP_FINANCIAL_TRANSACTION" ADD CONSTRAINT "SYS_C0078697" PRIMARY KEY
("FINANCIAL_TRANSACTION_SEQ", "REALM_ID")
USING INDEX PCTFREE 10 INITRANS 2 MAXTRANS 255 COMPUTE STATISTICS
TABLESPACE "PSP_IDX02" ENABLE

```

GoldenGate Configuration

Oracle Range Partitioning GG Extract

```

EXTRACT epspwsyf

SETENV (ORACLE_HOME="/u01/app/oracle/product/12.1.0.2")

-- Use this login when DB is source is local
USERIDALIAS ggsource_sys_monolith

-- Use this login when DB is source is remote
--USERIDALIAS ggsource_remote

TRANLOGOPTIONS EXCLUDETAG 2
--TRANLOGOPTIONS MINEFROMACTIVEDG

ENCRYPTTRAIL
EXTTRAIL ./dirdat/pspwsyf/tr

DDL INCLUDE MAPPED EXCLUDE UNMAPPED EXCLUDE OTHER
ddloptions GETREPLICATES, report

discardfile ./dirrpt/epspwsyf.dsc, megabytes 100, append
DISCARDROLLOVER AT 01:00

statoptions resetreportstats, reportfetch
report at 23:50
reportrollover at 00:01 on sunday

LOGALLSUPCOLS
--REPORTCOUNT EVERY 5 MINTUES RATE
UPDATERECORDFORMAT COMPACT

NOCOMPRESSDELETES
TABLE PSPADM.PSP_FINANCIAL_TRANSACTION;
SEQUENCE PSPADM.PSP_FINANCIAL_TRANSACTION;

```

Oracle Range Partitioning GG Replicat

```

REPLICAT rpspsyf

SETENV (ORACLE_HOME="/u01/app/oracle/product/12.1.0.2")
--SETENV (ORACLE_SID="psplpd01")

-- Use this login when DB is source is local
USERIDALIAS ggtarget_sys_monolith

DBOPTIONS SETTAG 2

-- Use this login when DB is source is remote
--USERIDALIAS ggtarget_remote

--BATCHSQL BATCHTRANSOPS 42
BATCHSQL

DBOPTIONS DEFERREFCONST
DBOPTIONS NOSKIPTEMPLOB

DBOPTIONS INTEGRATEDPARAMS (EAGER_SIZE 100000)

--DBOPTIONS INTEGRATEDPARAMS (parallelism 16)

DDL INCLUDE ALL EXCLUDE INSTR 'SYS_PLSQL' EXCLUDE INSTR 'ALTER INDEX'
DDLERROR 1435 IGNORE INCLUDE OPTYPE ALTER OBJTYPE SESSION

DISCARDFILE ./dirrpt/rpspsyf.dsc, APPEND, MEGABYTES 500
DISCARDROLLOVER at 00:01

--REPERROR 01403 DISCARD
--DDLERROR 942 IGNORE
--REPERROR (0001, discard)
--REPERROR 01400 DISCARD

statoptions resetreportstats
report at 23:58
reportrollover at 00:01

REPORTCOUNT EVERY 15 MINUTES, RATE
REPORTCOUNT EVERY 100000 RECORDS, RATE

APPLYNOOPUPDATES
ASSUMETARGETDEFS

MAP PSPADM.PSP_FINANCIAL_TRANSACTION, TARGET PSPADM_HASH.PSP_FINANCIAL_TRANSACTION;

```

SQLs

SQL commands tested to replication POC

```

insert into PSPADM.PSP_FINANCIAL_TRANSACTION (FINANCIAL_TRANSACTION_SEQ, VERSION, CREATOR_ID, CREATED_DATE,
MODIFIER_ID, MODIFIED_DATE, REALM_ID, TRANSACTION_TYPE_FK, COMPANY_FK)
values ('test', 100, 'test', sysdate, 'test', sysdate, -1, 'FLATemp1', '05ca91fc-53d1-4856-b6e1-a952c75720b1');
update PSPADM.PSP_FINANCIAL_TRANSACTION set FINANCIAL_TRANSACTION_AMOUNT=500 where
FINANCIAL_TRANSACTION_SEQ='test' and REALM_ID=-1;
delete from PSPADM.PSP_FINANCIAL_TRANSACTION where FINANCIAL_TRANSACTION_SEQ='test' and REALM_ID=-1;

```

Replication Observations

- DML(insert, update, delete) work without any additional configuration in replication.

Hash to Range Partitioning

Table DDL

```
CREATE TABLE "PSPADM_HASH"."PSP_FINANCIAL_TRANSACTION"
(
  "FINANCIAL_TRANSACTION_SEQ" VARCHAR2(255) NOT NULL ENABLE,
  "VERSION" NUMBER(19,0) NOT NULL ENABLE,
  "CREATOR_ID" VARCHAR2(30),
  "CREATED_DATE" TIMESTAMP (6) NOT NULL ENABLE,
  "MODIFIER_ID" VARCHAR2(30),
  "MODIFIED_DATE" TIMESTAMP (6) NOT NULL ENABLE,
  "REALM_ID" NUMBER(19,0) DEFAULT -1 NOT NULL ENABLE,
  "FINANCIAL_TRANSACTION_AMOUNT" NUMBER(19,4),
  "SETTLEMENT_DATE" TIMESTAMP (6),
  "SETTLEMENT_TYPE_CD" VARCHAR2(255),
  "CREDIT_BANK_ACCOUNT_TYPE" VARCHAR2(255),
  "DEBIT_BANK_ACCOUNT_TYPE" VARCHAR2(255),
  "ON_HOLD" NUMBER(1,0),
  "SKU" VARCHAR2(40),
  "SKU_QUANTITY" NUMBER(10,0),
  "BILLING_DETAIL_FK" VARCHAR2(255),
  "CREDIT_BANK_ACCOUNT_FK" VARCHAR2(255),
  "DEBIT_BANK_ACCOUNT_FK" VARCHAR2(255),
  "COMPANY_FK" VARCHAR2(255),
  "PAYROLL_RUN_FK" VARCHAR2(255),
  "PAYCHECK_SPLIT_FK" VARCHAR2(255),
  "TRANSACTION_TYPE_FK" VARCHAR2(255) NOT NULL ENABLE,
  "LAW_FK" VARCHAR2(255),
  "CURRENT_TRANSACTION_STATE_FK" VARCHAR2(255),
  "ORIGINAL_TRANSACTION_FK" VARCHAR2(255),
  "MONEY_MOVEMENT_TRANSACTION_FK" VARCHAR2(255),
  "ORIGINAL_SETTLEMENT_DATE" TIMESTAMP (6),
  "REFUND_TYPE" VARCHAR2(255 CHAR),
  "COMP_ADJUST_SUBMISSION_FK" VARCHAR2(255 CHAR),
  "TAX_PENALTY_INTEREST_FK" VARCHAR2(255 CHAR),
  "RELATABLE_TRANSACTION_FK" VARCHAR2(255 CHAR),
  "BILL_PAYMENT_SPLIT_FK" VARCHAR2(255 CHAR),
  "COMPANY_LAW_FK" VARCHAR2(255 CHAR),
  "STATUS" VARCHAR2(255 CHAR),
  CONSTRAINT "PSP_FINANCIAL_TRANSACTION_PK" PRIMARY KEY ("COMPANY_FK", "FINANCIAL_TRANSACTION_SEQ")
USING INDEX PCTFREE 10 INITRANS 2 MAXTRANS 255 COMPUTE STATISTICS
TABLESPACE "PSP_IDX02" ENABLE
) PCTFREE 10 PCTUSED 40 INITRANS 1 MAXTRANS 255
TABLESPACE "PSP_DATA01"
PARTITION BY HASH ("COMPANY_FK")
(PARTITION "SYS_P15840" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15841" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15842" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15843" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15844" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15845" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15846" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15847" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15848" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15849" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15850" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15851" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15852" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15853" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15854" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15855" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15856" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15857" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15858" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15859" TABLESPACE "PSP_DATA01",
```

```

PARTITION "SYS_P15860" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15861" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15862" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15863" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15864" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15865" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15866" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15867" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15868" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15869" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15870" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15871" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15872" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15873" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15874" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15875" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15876" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15877" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15878" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15879" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15880" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15881" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15882" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15883" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15884" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15885" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15886" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15887" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15888" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15889" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15890" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15891" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15892" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15893" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15894" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15895" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15896" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15897" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15898" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15899" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15900" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15901" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15902" TABLESPACE "PSP_DATA01",
PARTITION "SYS_P15903" TABLESPACE "PSP_DATA01")

```

GoldenGate Replication

Oracle Range Partitioning GG Extract

```

EXTRACT epsphsyf

SETENV (ORACLE_HOME="/u01/app/oracle/product/12.1.0.2")

-- Use this login when DB is source is local
USERIDALIAS ggsource_sys_monolith

-- Use this login when DB is source is remote
--USERIDALIAS ggsource_remote

TRANLOGOPTIONS EXCLUDETAG 2
--TRANLOGOPTIONS MINEFROMACTIVEDG

ENCRYPTTRAIL
EXTTRAIL ./dirdat/psphsyf/tr

DDL INCLUDE MAPPED EXCLUDE UNMAPPED EXCLUDE OTHER
ddloptions GETREPLICATES, report

discardfile ./dirrpt/epsphsyf.dsc, megabytes 100, append
DISCARDROLLOVER AT 01:00

statoptions resetreportstats, reportfetch
report at 23:50
reportrollover at 00:01 on sunday

LOGALLSUPCOLS
--REPORTCOUNT EVERY 5 MINTUES RATE
UPDATERECORDFORMAT COMPACT

NOCOMPRESSDELETES
TABLE PSPADM_HASH.PSP_FINANCIAL_TRANSACTION;
SEQUENCE PSPADM_HASH.PSP_FINANCIAL_TRANSACTION;

```

Oracle Range Partitioning GG Replicat

```

REPLICAT rpsphsyf

SETENV (ORACLE_HOME="/u01/app/oracle/product/12.1.0.2")
--SETENV (ORACLE_SID="psplpd01")

-- Use this login when DB is source is local
USERIDALIAS ggtarget_sys_monolith

DBOPTIONS SETTAG 2

-- Use this login when DB is source is remote
--USERIDALIAS ggtarget_remote

--BATCHSQL BATCHTRANSOPS 42
BATCHSQL

DBOPTIONS DEFERREFCONST
DBOPTIONS NOSKIPTEMPLOB

DBOPTIONS INTEGRATEDPARAMS (EAGER_SIZE 100000)

--DBOPTIONS INTEGRATEDPARAMS (parallelism 16)

DDL INCLUDE ALL EXCLUDE INSTR 'SYS_PLSQL' EXCLUDE INSTR 'ALTER INDEX'
DDLERROR 1435 IGNORE INCLUDE OPTYPE ALTER OBJTYPE SESSION

DISCARDFILE ./dirrpt/rpsphsyf.dsc, APPEND, MEGABYTES 500
DISCARDROLLOVER at 00:01

--REPERROR 01403 DISCARD
--DDLERROR 942 IGNORE
--REPERROR (0001, discard)
--REPERROR 01400 DISCARD

statoptions resetreportstats
report at 23:58
reportrollover at 00:01

REPORTCOUNT EVERY 15 MINUTES, RATE
REPORTCOUNT EVERY 100000 RECORDS, RATE

APPLYNOOPUPDATES
ASSUMETARGETDEFS

MAP PSPADM_HASH.PSP_FINANCIAL_TRANSACTION, TARGET PSPADM.PSP_FINANCIAL_TRANSACTION, KEYCOLS (COMPANY_FK,
FINANCIAL_TRANSACTION_SEQ);
MAP PSPADM_HASH.PSP_FINANCIAL_TRANSACTION, TARGET PSPADM.PSP_FINANCIAL_TRANSACTION;

```

SQLs

```

insert into PSPADM_HASH.PSP_FINANCIAL_TRANSACTION (FINANCIAL_TRANSACTION_SEQ, VERSION, CREATOR_ID,
CREATED_DATE, MODIFIER_ID, MODIFIED_DATE, REALM_ID, TRANSACTION_TYPE_FK, COMPANY_FK)
values ('test', 100, 'test', sysdate, 'test', sysdate, -1, 'FLATemp1', '05ca91fc-53d1-4856-b6e1-a952c75720b1');
update PSPADM_HASH.PSP_FINANCIAL_TRANSACTION set FINANCIAL_TRANSACTION_AMOUNT=500 where COMPANY_FK='05ca91fc-
53d1-4856-b6e1-a952c75720b1' and FINANCIAL_TRANSACTION_SEQ='test';
delete from PSPADM_HASH.PSP_FINANCIAL_TRANSACTION where COMPANY_FK='05ca91fc-53d1-4856-b6e1-a952c75720b1' and
FINANCIAL_TRANSACTION_SEQ='test';

```


Replication Observations

- Insert, Delete work without any additional configuration in replication.
- For Update, we need to add KEYCOLS parameter in replicate configuration.